lowest volume, as you can see also for a 2D space in figure 10.23: the purple circle of radius `r` is inscribed in a square of radius `2r`; if the points are distributed in a circular cluster, then all the areas (highlighted in gray) of the square that does not intersect the circle are potentially empty, and so wasted.

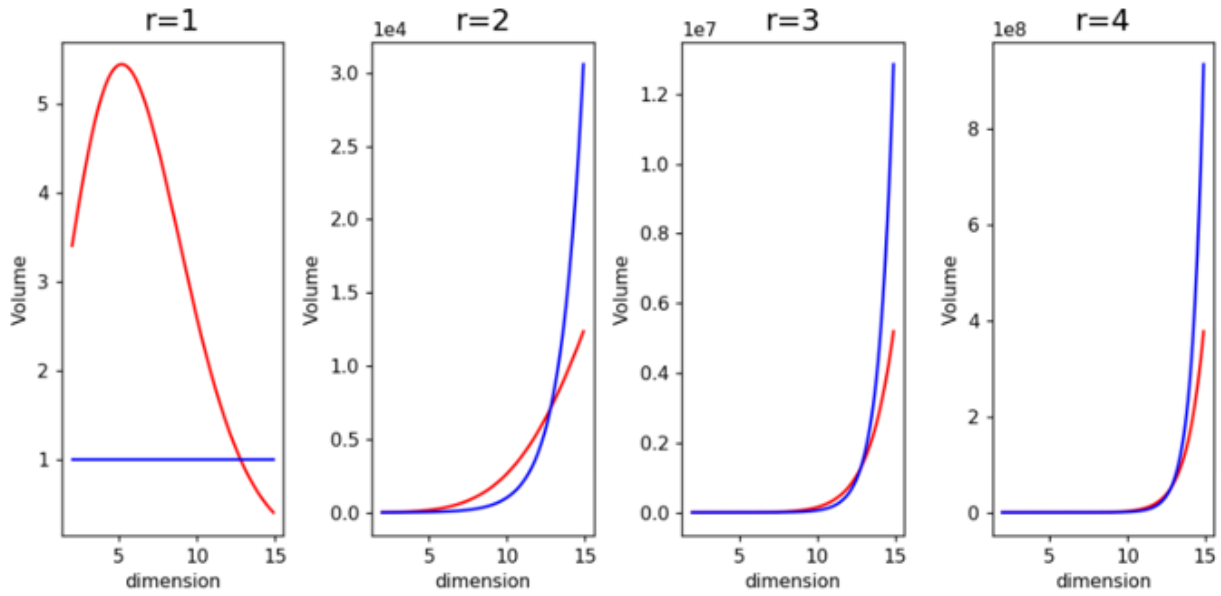


Figure 10.25 Volume of spheres (red) and cubes (blue) for various radii, as a function of the number os dimensions of the search space.

Experiments have confirmed that Ss-trees using spherical bounding envelopes performs better on datasets uniformly distributed along all directions, while rectangular envelopes work better with skewed datasets.

Table 10.1 Operations provided by k-d tree, and their cost on a balanced k-d